Modeling the First COVID-19 Wave in Hubei with a Simple SIR Model

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Abstract

This project uses a simple susceptible-infected-removed (SIR) model to study the first reported COVID-19 wave in Hubei, China. Using Johns Hopkins CSSE cumulative confirmed case data from January 22, 2020 to April 30, 2020, I fit the model to reported cumulative cases and then use the fitted model to test hypothetical reductions in the transmission rate. The fitted model gives a transmission rate $\beta \approx 0.329$, removal rate $\gamma \approx 0.065$, and effective population size $N_{\text{eff}} \approx 68{,}809$. In the counterfactual simulations, reducing β lowers the peak number of model daily cases and shifts the peak later in time. These results support the general "flattening the curve" idea, while also showing the limitations of applying a constant-parameter SIR model to reported COVID-19 case data.

1 Introduction

Compartment models are a common way to study infectious disease dynamics. In an SIR model, the population is divided into susceptible, infected, and removed individuals, and the flow between these groups is described by differential equations. Cooper, Mondal, and Antonopoulos used an SIR-type model to study COVID-19 spread in different communities, showing how a relatively simple model can still give insight into epidemic growth and control [2].

The scientific question in this project is:

> Can a simple SIR model approximate the first COVID-19 wave in Hubei, and how would reducing the transmission rate affect the infection peak and total outbreak size?

I focus on Hubei because it was the center of the earliest large COVID-19 outbreak in China and because province-level time series data are available in a clean form. The goal is not to build a complete COVID-19 forecasting model. Instead, the goal is to test how much can be learned from a basic SIR model and to interpret where that model is useful or too simple.

2 Data

The data come from the Johns Hopkins CSSE COVID-19 time series repository [3]. I use the global confirmed cases file and select the row with province/state equal to Hubei and country/region equal to China. The time period is January 22, 2020 through April 30, 2020, which captures the first reported wave in Hubei.

The original dataset reports cumulative confirmed cases by date. I also compute daily new cases by differencing the cumulative counts. Figure 1 shows the reported daily new cases. One noticeable feature is the large reporting jump in mid-February 2020. This is important because the model is smooth, but reported data can change suddenly due to testing capacity or case definition changes.

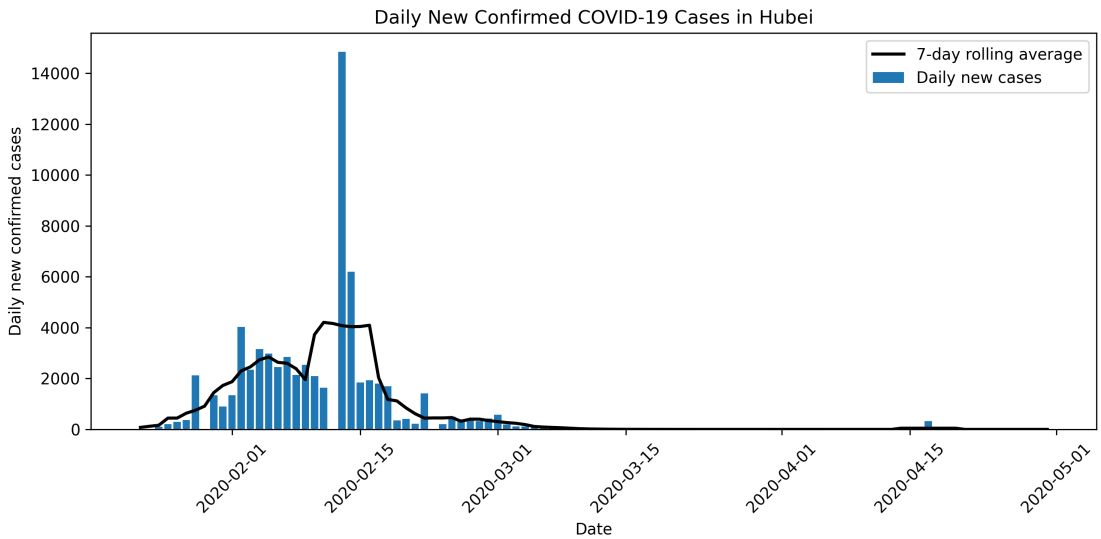

Figure 1: Daily new confirmed COVID-19 cases in Hubei from January 22, 2020 to April 30, 2020. The black curve shows a 7-day rolling average.

3 Model

The baseline model is the standard SIR system

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \tag{1}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \tag{2}$$

$$\frac{dR}{dt} = \gamma I. \tag{3}$$

Here $S(t)$ is the susceptible population, $I(t)$ is the infected population, $R(t)$ is the removed or recovered population, and N is the effective population size used in the model. The parameter β is the transmission rate, and γ is the removal rate. The basic reproduction number in this simple model is

$$R_0 = \frac{\beta}{\gamma}. \tag{4}$$

The cumulative number of infections generated by the model is

$$C(t) = N - S(t). \tag{5}$$

I fit this model-generated cumulative case count to the reported cumulative confirmed cases.

4 Methods

I solved the SIR differential equations numerically using Python's `solve_ivp`. The fitted parameters were β, γ, and an effective population size N_{eff}. I used least squares optimization to minimize the scaled difference between the model cumulative cases and the reported cumulative confirmed cases.

After fitting the baseline model, I ran four transmission scenarios:

$$\beta, \quad 0.9\beta, \quad 0.8\beta, \quad 0.7\beta.$$

These represent the fitted transmission rate and hypothetical 10%, 20%, and 30% reductions in transmission. For each scenario, I recorded the peak model daily cases, the date of the peak, and the final cumulative cases over the study period.

5 Results

The fitted parameter values were

$$\beta = 0.32884, \qquad \gamma = 0.06530, \qquad R_0 = 5.04, \qquad N_{\text{eff}} = 68{,}809.$$

The value of N_{eff} should not be interpreted as the full population of Hubei. It is an effective population size for the reported first-wave outbreak under this simplified model.

Figure 2 compares the fitted SIR cumulative case curve with the reported cumulative confirmed cases. The model captures the broad S-shaped pattern of the first wave, but it does not capture abrupt reporting changes exactly.

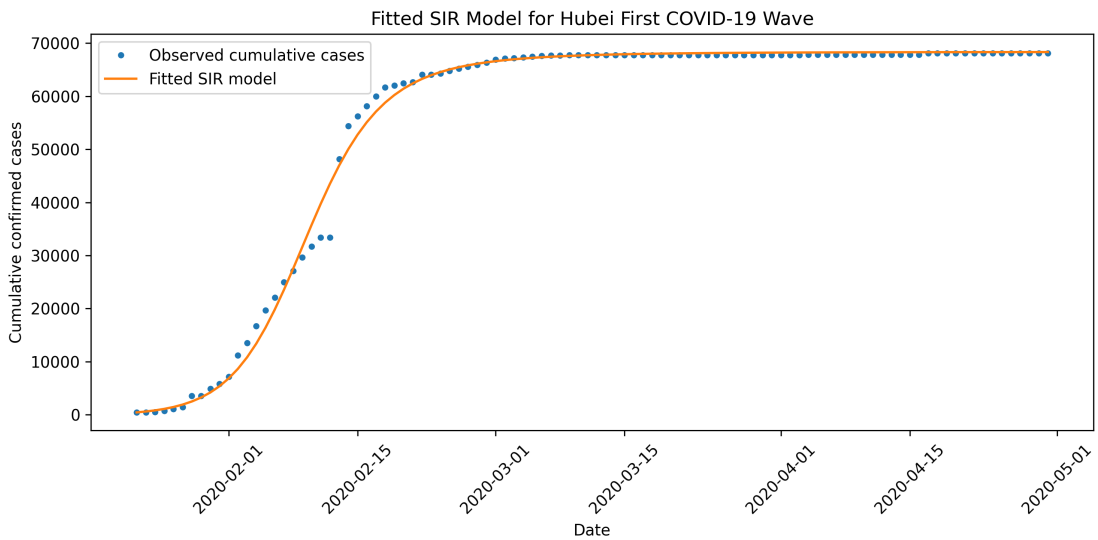

Figure 2: Fitted SIR model compared with observed cumulative confirmed cases in Hubei.

The counterfactual simulations show that lowering the transmission rate reduces the peak number of model daily cases and delays the peak. Figure 3 shows the daily case curves under each transmission scenario.

The most visible effect is on the peak. A 30% reduction in β lowers the peak from about 4,141 model daily cases to about 2,460 model daily cases, a decrease of about 41%. The peak also shifts from February 9 to February 19. In this fixed study window, the final cumulative case count also decreases, although less dramatically than the peak.

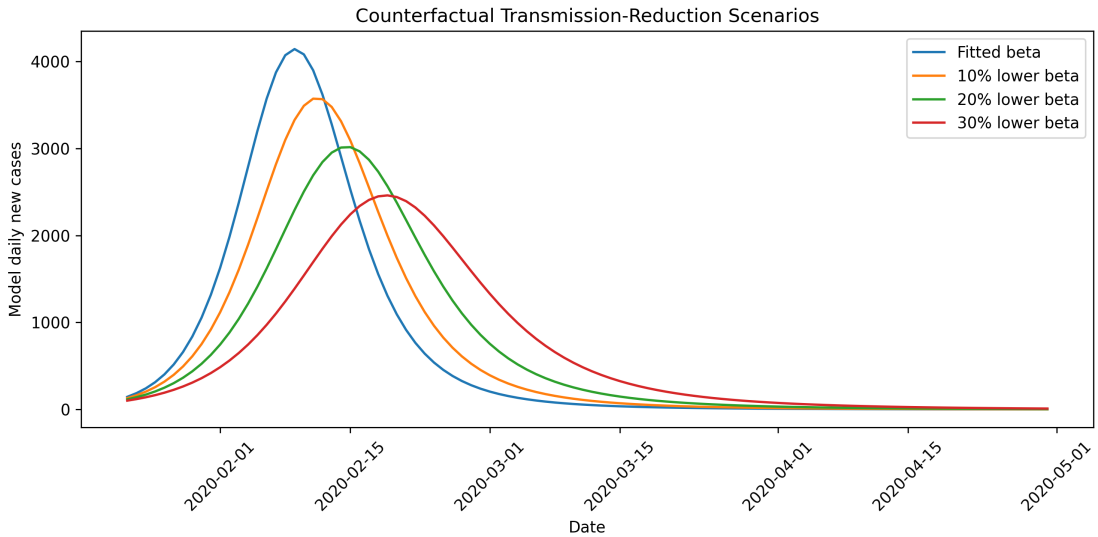

Figure 3: Counterfactual SIR simulations with the fitted transmission rate and 10%, 20%, and 30% lower transmission rates.

Scenario	Peak daily cases	Peak date	Final cumulative cases
Fitted β	4,141	2020-02-09	68,332
10% lower β	3,571	2020-02-11	68,001
20% lower β	3,014	2020-02-15	67,421
30% lower β	2,460	2020-02-19	66,369

Table 1: Summary of transmission-reduction scenarios.

6 Discussion and Limitations

The model supports the idea that reducing transmission can flatten an epidemic curve. In the simulations, lower values of β produce lower and later peaks. This is consistent with the public health interpretation of interventions that reduce contact rates or transmission probability.

However, this model has several limitations. First, the SIR model assumes homogeneous mixing, so every susceptible individual is treated as equally likely to contact every infected individual. Real COVID-19 spread depends on household structure, age, occupation, travel, and local policy. Second, reported confirmed cases are not the same as true infections. They depend on testing availability, reporting rules, and case definitions. Third, the fitted model uses constant β and γ, while real transmission changed over time because of behavior changes and interventions. Finally, the model does not include exposed but not yet infectious individuals, asymptomatic infection, hospitalization, deaths as a separate compartment, or outside infections. AlQadi and Bani-Yaghoub discuss how adding global dynamics and external infection effects can improve COVID-19 SIR-type models [1], which highlights why the standard SIR model should be interpreted cautiously.

Overall, the simple SIR model is useful for approximating the broad shape of the first Hubei wave and for testing basic transmission-reduction scenarios. It is less useful as a detailed forecasting model for real policy decisions.

7 Conclusion

A simple SIR model can approximate the broad cumulative case pattern of the first reported COVID-19 wave in Hubei. After fitting the model to reported cumulative cases, reducing the transmission rate in counterfactual simulations lowered the daily case peak and delayed its timing. The project shows how a basic mathematical model can connect real data to a scientific question, while also making clear that reported COVID-19 dynamics are more complicated than the assumptions of a standard SIR system.

References

[1] Hadeel AlQadi and Majid Bani-Yaghoub. Incorporating global dynamics to improve the accuracy of disease models: Example of a covid-19 sir model. *PLOS ONE*, 17(4):e0265815, 2022. doi: 10.1371/journal.pone.0265815.

[2] Ian Cooper, Argha Mondal, and Chris G. Antonopoulos. A sir model assumption for the spread of covid-19 in different communities. *Chaos, Solitons & Fractals*, 139:110057, 2020. doi: 10.1016/j.chaos.2020.110057.

[3] Johns Hopkins University Center for Systems Science and Engineering. Covid-19 data repository. `https://github.com/CSSEGISandData/COVID-19`, 2020. Accessed for historical confirmed case time series data.